C 40605 (Pages : 4) Name.....

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS—UG)

Mathematics

MTS 6B 12—CALCULUS OF MULTI VARIABLE

(2019 Admission onwards)

Time : Two Hours and a Half Maximum : 80 Marks

Section A

Questions 1–15*, Answer any number of questions.*
Each question carries 2 *marks.*
Maximum marks 25.

1. Find the domain and range of the function $f(x, y) = \sqrt{4 - x^2 - y^2}$.

2. Show that $\lim_{(x, y) \to (0, 0)} \frac{xy^2}{x^2 + y^4}$ does not exist.

3. Find f_x and f_y if $f(x, y) = y^x$.

4. Find the Directional Derivative of $f(x, y) = 4 - x^2 - \frac{y^2}{4}$ at $(1, 2)$ in the direction of $\vec{u} = \cos\frac{\pi}{3}\hat{i} + \sin\frac{\pi}{3}\hat{j}$.

5. Find the gradient of $f(x, y) = y \ln x + xy^2$ at the point $(1, 2)$.

6. Find the relative extrema of $f(x, y) = 1 - \left(x^2 + y^2\right)^{1/3}$.

Turn over

7. Evaluate $\iint_R 2x - y \, dA$ where R is the region bounded by the parabola $x = y^2$ and the straight line $x - y = 2$.

8. Find the surface area of the portion of the plane $z = 2 - x - y$ that lies above the circle $x^2 + y^2 \leq 1$ in the first quadrant.

9. Find a transformation T from a region S to the region R bounded by the lines $x - 2y = 0, x - 2y = -4, x + y = 4$ and $x + y = 1$ such that S is a rectangular region.

10. Sketch the region of integration and reverse the order of integration in $\int_1^e \int_0^{\ln x} f(x, y) \, dy \, dx$.

11. Check whether the vector field $\vec{F}(x, y, z) = 2xy \, \hat{i} + \left(x^2 + z^2\right) \hat{j} + 2yz \, \hat{k}$ is irrotational.

12. Find div $\left(\text{curl } \vec{F}\right)$ if $\vec{F}(x, y, z) = xyz \, \hat{i} + y\hat{j} + z\hat{k}$.

13. Use Green's theorem to evaluate $\oint_C \left(x^2 y + x^3\right) dx + 2xy \, dy$ where C if the boundary of the region bounded by $y = x$ and $y = x^2$.

14. State Gauss Divergence theorem.

15. Evaluate the surface integral $\iint_S x + z \, dS$ where S is the first octant portion of the cylinder $y^2 + z^2 = 9$ between $x = 0$ and $x = 4$.

Section B

Questions 16–23, *Answer any number of questions.*
Each question carries 5 *marks.*
Maximum marks 35.

16. Show that the function $z = \sqrt{x^2 + y^2}\ \tan^{-1}\frac{y}{x}$ satisfies $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z$.

17. Let $w = x^2y - xy^3$ where $x = \cos t$ and $y = e^t$. Find $\frac{dw}{dt}$ at $t = 0$.

18. Find the points on the sphere $x^2 + y^2 + z^2 = 14$ at which the tangent plane is parallel to the plane $x + 2y + 3z = 12$.

19. Find the relative extrema of $f(x, y) = 2x^2 + y^2 + 8x - 6y + 20$.

20. Find the volume of the solid bounded by the cylinder $y^2 + z^2 = 9$ and the planes $x = 0, y = 0, z = 0$ and $2x + y = 2$.

21. Evaluate $\int_0^1 \int_y^1 \frac{\sin x}{x}\, dx\, dy$ by changing the order of integration.

22. Find an equation of the tangent plane to the paraboloid $\vec{r}(u, v) = u\hat{i} + v\hat{j} + \left(u^2 + v^2\right)\hat{k}$ at the point $(1, 2, 5)$.

23. Find the surface integral $\iint_S y^2 + 2yz\, d\mathrm{S}$ where S is the first octant portion of the plane $2x + y + 2z = 6$.

Section C

Questions 24–27, *Answer any* **two** *questions.*
Each question carries 10 *marks.*

24. (a) Find the second order partial derivatives of $w = \cos(2u - v) + \sin(2u + v)$.

(b) Let $z = f(x, y) = 2x^2 - xy$. Find Δz and use the result to find the change in z if (x, y) changes from $(1, 1)$ to $(0.98, 1.03)$.

25. Find the absolute maximum and absolute minimum values of the function $f(x, y) = 2x^2 + y^2 - 4x - 2y + 3$ on the rectangle $D = \{(x, y) \mid 0 \le x \le 3, 0 \le y \le 2\}$.

26. Evaluate $\iiint_T \sqrt{x^2 + z^2}\, dv$ where T is the region bounded by the cylinder $x^2 + z^2 = 1$ and the planes $y + z = 2$ and $y = 0$.

27. verify Divergence Theorem for $\vec{F}(x, y, z) = 2z\hat{i} + x\hat{j} + y^2\hat{k}$ and T is the solid region bounded between the paraboloid $z = 4 - x^2 - y^2$ and the xy plane.

(2 × 10 = 20 marks)

C 40605–A (Pages : 6) Name.....

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SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS—U.G.)

Mathematics

MTS 6B 12—CALCULUS OF MULTI VARIABLE

(2019 Admission onwards)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	Total No. of Questions : 20	Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 12—CALCULUS OF MULTI VARIABLE

(Multiple Choice Questions for SDE Candidates)

1. Evaluate $\iiint_T x dV$, Where T is the part of the region in the first octant lying inside the sphere $x^2 + y^2 + z^2 = 1$.

   (A) $\frac{\pi}{6}$. (B) $\frac{\pi}{16}$.

   (C) $\frac{\pi}{4}$. (D) π.

2. Evaluate $\lim_{(x, y) \to (2, 4)} \sqrt[3]{\frac{8xy}{2x + y}}$.

   (A) 2. (B) 3.

   (C) – 2. (D) – 3.

3. Determine where the function is continuous : $f(x, y) = \frac{xy\left(x^2 - y^2\right)}{x^2 + y^2}$.

   (A) Everywhere except at (0,0). (B) Everywhere.

   (C) Only at the center. (D) At (1, 1).

4. Determine where the function G $(x, y) = \frac{\frac{1}{2}\cos\left(2x^2 + y^2\right)}{1 + 2x^2 + y^2}$ is continuous :

   (A) Everywhere. (B) At (1, 1).

   (C) At (0, 0). (D) None of these.

5. Find $\frac{\partial f}{\partial x}$ if $f(x, y) = 2x^2 y^3 - 3xy^2 + 2x^2 + 3y^2 + 1$.

   (A) $4xy^3 - 3y^2 + 4x$. (B) $4x^2y^3$.

   (C) $6x^2 y^3 - 6xy^2 + 4x^2 + 3y^2$. (D) $6x^2 y^3 - 6xy^2 + 2x^2 + 3y^2$.

6. Find f_y if $f(x, y) = x \cos xy^2$.

(A) $-2x^2 y \sin xy^2$. (B) $2x^2 \sin xy^2$.

(C) $2x^2 y \sin xy^2$. (D) 0.

7. Find the second-order partial derivatives of $f(x, y) = 2xy^2 - 3x^2 + xy^3$.

(A) $4x + 6xy$. (B) $6xy$.

(C) $2x + 6xy$. (D) 0.

8. The function f defined by $f(x, y) = 2x^2 - xy$ is ———— in the plane.

(A) Differentiable. (B) Not-differentiable.

(C) Discontinuous. (D) None of these.

9. Find $\frac{dy}{dx}$ if $x^3 + xy + y^2 = 4$.

(A) $-\frac{3x^2 + y}{x + 2y}$. (B) $-\frac{3x + y}{x + 2y}$.

(C) $\frac{3x^2 + y}{x + 2y}$. (D) $-\frac{3x^2 + y}{x + y}$.

10. Find the directional derivative of $f(x, y) = e^x \cos 2y$ at the point $\left(0, \frac{\pi}{4}\right)$ in the direction of $\mathbf{v} = \mathbf{2i} + \mathbf{3j}$.

(A) $-\frac{6\sqrt{13}}{13}$. (B) 1.

(C) 0. (D) $\frac{6\sqrt{13}}{13}$.

11. Find the area enclosed by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

(A) $(\pi ab)^2$. (B) $\pi a^2 b$.

(C) $\pi^2 ab$ (D) πab.

12. Find the flux of the vector field $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j} + 2z\mathbf{k}$ across the unit sphere $x^2 + y^2 + z^2 = 1$.

(A) $\frac{8\pi}{6}$. (B) $\frac{8\pi^2}{3}$.

(C) $\frac{8\pi}{3}$. (D) $\frac{8\pi}{2}$.

13. Find the formula for Divergence Theorem :

(A) $\iint_s \mathbf{F}\cdot d\mathbf{S} = \iiint_T div\ \mathbf{F} dV$. (B) $\iint_s \mathbf{F}\cdot d\mathbf{S} = -\iiint_T div\ \mathbf{F} dV$.

(C) $\iint_s \mathbf{F}\cdot d\mathbf{S} = \iint_T div\ \mathbf{F} dV$. (D) $\int_s \mathbf{F}\cdot d\mathbf{S} = \iiint_T div\ \mathbf{F} dV$.

14. Let f be defined on an open region R containing the point (a, b). We call (a, b) a critical point of f if :

A f_x and /or f_y do not exist at (a, b) or.

B Both $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

(A) A is wrong. (B) B is wrong.

(C) Both A and B are Correct. (D) None of these.

15. Let f be continuous on a polar rectangle $R = \{(r, \theta) | 0 \le a \le r \le b, \alpha \le \theta \le \beta\}$, where $0 \le \beta - \alpha \le 2\pi$. Then $\iint_R f(x, y)\, dA =$

(A) $\int_\alpha^\beta \int_a^b f(\cos\theta, \sin\theta)\, r\, dr\, d\theta$.

(B) $\int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, dr\, d\theta$.

(C) $\int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$.

(D) $-\int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$.

16. Let f be continuous on the rectangular box $B = \{(x, y, z) | a \le x \le b, c \le y \le d, p \le z \le q\}$. Then $\iiint_B f(x, y, z)\, dV =$

(A) $\int_c^d \int_a^b f(x, y, z)\, dxdydz$.

(B) $\int_p^q \int_c^d \int_a^b f(x, y, z)\, dxdydz$.

(C) $\int_p^c \int_c^d \int_a^b f(x, y, z)\, dxdydz$.

(D) $\int_p^q \int_c^a \int_a^b f(x, y, z)\, dxdydz$.

17. The derivatives f_{xy} and f_{yx} are called ————.

(A) Partial derivatives

(B) Mixed derivatives.

(C) Mixed partial derivatives.

(D) None of these

18. Let f be a function of two variables x and y. The gradient of f is the vector function $\nabla f(x, y) =$

(A) $f_y(x, y)\,\mathbf{i} + f_x(x, y)\,\mathbf{j}$.

(B) $f_x(x, y)\,\mathbf{j} + f_y(x, y)\,\mathbf{i}$.

(C) $f_x(x, y)\,\mathbf{i} + f_y(x, y)\,\mathbf{j}$.

(D) $f_x(x, y)\,\mathbf{i} + f_y(x, y)\,\mathbf{k}$.

19. Suppose that f has continuous second-order partial derivatives on an open region containing a critical point (a, b) of f. Let $D(x, y) = f_{xx}(x, y) f_{yy}(x, y) - f_{xy}^2(x, y)$. If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a ————.

(A) Relative minimum value. (B) Saddle point.

(C) Relative maximum value. (D) None of these.

20. if f is continuous on a closed, bounded set D in the plane, then f attains a/an at some point (a, b) in D :

(A) Absolute maximum value. (B) Absolute minimum value.

(C) Minimum value. (D) None of these.